
Software Requirements Specification

For

**An informatic system for the management of a medical
clinic**

Version 4.0 approved

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Revision History

Name	Date	Reason For Changes	Version
Ungureanu Lorena-Mihaela	02.03.2026	Adding 1.1, 1.2, 1.5, 5.1, 5.2, 5.3, 5.4, 5.5	1.0

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1. Introduction

1.1 Purpose

The purpose of this application is to facilitate the efficient management of a medical clinic by automating medical, administrative, and scheduling processes, in order to improve service quality, resource utilization, and communication between patients and medical staff.

1.2 Document Conventions

Bold text is used to highlight section headings and important terms throughout the document.

1.3 Intended Audience and Reading Suggestions

This Software Requirements Specification (SRS) document outlines the architecture, functionalities, and constraints of a medical clinic management system designed to support multiple medical specialties. To maximize efficiency, readers are encouraged to review the sections most pertinent to their specific roles:

- *Developers should focus on the technical implementation, architecture, and coding logic required to build the core modules. They are responsible for programming the dependencies between user roles (doctor, nurse, administrator, patient) , configuring the calendar allocation system , and implementing the dual rating system for doctors and patients.*
 - *Key sections for them include:*
 - **Product Scope (1.4)**
 - **Product Perspective (2.1)**
 - **Operating Environment (2.4)**
 - **Product Functions (2.2)**
 - **External Interface Requirements (3)**
 - **System Features (4)**
 - **Other Nonfunctional Requirements (5)**
- *Project Managers require a high-level understanding of the project's workload, deliverables, and operational constraints. They must oversee complex business rules, such as enforcing the mandatory presence of a nurse during procedures or restricting specific rooms to certain medical specializations.*
 - *They should refer to:*
 - **Product Scope (1.4)**
 - **Product Perspective (2.1)**
 - **Design and Implementation Constraints (2.5)**
 - **Document Conventions (1.2)**
 - **Operating Environment (2.4)**
 - **Product Functions (2.2)**
- *Testers must understand the expected behavior of all features to design effective test cases and ensure system stability. They will validate critical workflows, such as the creation and editing of appointments (by patients or nurses/secretaries) , the toggling of resource*

availability (e.g., placing a device in maintenance mode) , and ensuring patient files are accessible to both doctors and patients.

- The key sections for them are:
 - *Product Scope (1.4)*
 - *Product Perspective (2.1)*
 - *Product Functions (2.2)*
 - *System Features (4)*
- *Documentation Writers need to fully comprehend the product's purpose and user interfaces to draft accurate manuals and help materials. They will document how users can define resources , how to track a patient's evolution based on investigations , and how to add or edit external medical records (past treatments, external analyses).*
 - They should refer to:
 - *Product Scope (1.4)*
 - *Product Perspective (2.1)*
 - *Purpose (1.1)*
 - *Document Conventions (1.2)*
 - *References (1.5)*
 - *Other Requirements (6)*

1.4 Product Scope

The primary objective of the medical clinic management system is to streamline and automate the daily operations of a clinic that encompasses multiple medical specialties. The software is designed to define, manage, and allocate various resources, such as medical cabinets, diagnostic devices, and procedure rooms, to its users, including patients, doctors, nurses, and maintenance staff. Furthermore, it handles appointment scheduling and maintains comprehensive patient medical records.

By implementing this system, the clinic will benefit from:

- **Optimized resource management:** *Efficient allocation of medical rooms and equipment through a centralized calendar system.*
- **Enforced operational constraints:** *Automatic enforcement of rules, such as restricting specific rooms to certain medical specializations or requiring the presence of a nurse during specific procedures.*
- **Improved accessibility and patient experience:** *Enabling patients to book online appointments and view their own medical files.*
- **Comprehensive medical tracking:** *Allowing doctors to visualize a patient's evolution based on internal analyses and integrate external medical history, such as previous treatments or external investigations.*
- **Increased accountability:** *Fostering a high-quality service environment through a rating system.*

1.5 References

- <https://www.reginamaria.ro/>
- <https://www.medlife.ro/rezultate-analize>
- <https://www.sanador.ro/>
- <https://www.mayoclinic.org/>

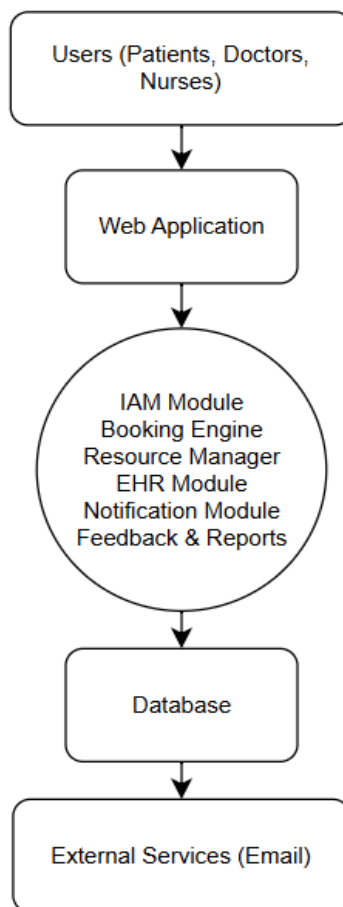
2. Overall Description

2.1 Product Perspective

*The **Clinic Management System** is a new web-based application designed to support the digital management of a medical clinic. It replaces manual processes and fragmented tools used for managing appointments, patient records, and clinic resources.*

The system integrates multiple clinic operations into a single platform where patients, doctors, and administrative staff can interact through a secure web interface. It manages user accounts, appointment scheduling, medical records, clinic resources, notifications, and feedback.

The application follows a modular architecture composed of several interconnected components.



2.2 Product Functions

The Medical Clinic Management System provides role-based access and functionalities tailored to different user categories to manage a clinic encompassing multiple specialties. The main user categories include Admin, Doctor, Nurse/Secretary, and Patient.

Authentication & User Roles

- *Users can access the platform based on predefined core roles such as medic, nurse, administrator, and patient.*
- *The platform manages the specific dependencies and interactions between these users.*

Patient Functionalities

- *Make online appointments directly through the platform.*
- *View their personal patient medical file.*
- *Participate in a rating system by rating doctors.*

Doctor Functionalities

- *View the medical files of their patients.*
- *Add, edit, or delete external resources regarding patients, such as investigations, analyses, and treatments received previously or from other facilities.*
- *Visualize a patient's medical evolution based on conducted analyses and investigations.*
- *Participate in the rating system by rating patients and receiving ratings.*

Nurse / Secretary Functionalities

- *Create appointments on behalf of patients.*
- *Edit existing patient appointments.*
- *Be allocated to specific consultations or procedures where the presence of a nurse is a mandatory constraint.*

Admin Functionalities

- *Define basic user roles and the dependencies between them.*
- *Define and name clinic resources, including medical cabinets, diagnostic devices, analysis devices, and procedure rooms.*
- *Define complex restrictions and dependencies, such as restricting certain rooms strictly to a specific medical specialization.*
- *Manage a centralized calendar where clinic resources are allocated for appointments.*
- *Activate or deactivate resources based on availability, such as marking a room unavailable for renovation or a device unavailable for maintenance and service.*

2.3 User Classes and Characteristics

The medical clinic management system consists of several core entities, listed below in order of their importance within the application:

1. Administrator Entity

- *Properties:*
 - *id (Unique identifier)*

- *name (Full name)*
 - *email (Contact email)*
- *Role: Responsible for defining basic roles such as doctor, nurse, administrator, and patient, along with the dependencies between them. They define resources and their names and configure complex restrictions and dependencies , such as allocating certain rooms strictly for a specific medical specialization.*

2. Doctor Entity

- *Properties:*
 - *id (Unique identifier)*
 - *name (Full name)*
 - *specialization (Medical specialty)*
 - *rating (Score from the system)*
 - *list of patients (Assigned patients)*
- *Role: Views the patient's medical file and visualizes a patient's evolution based on performed analyses and investigations. They can add, edit, or delete external resources regarding patients, such as investigations, analyses, or prior treatments received elsewhere. They are also part of a rating system.*

3. Nurse / Secretary Entity

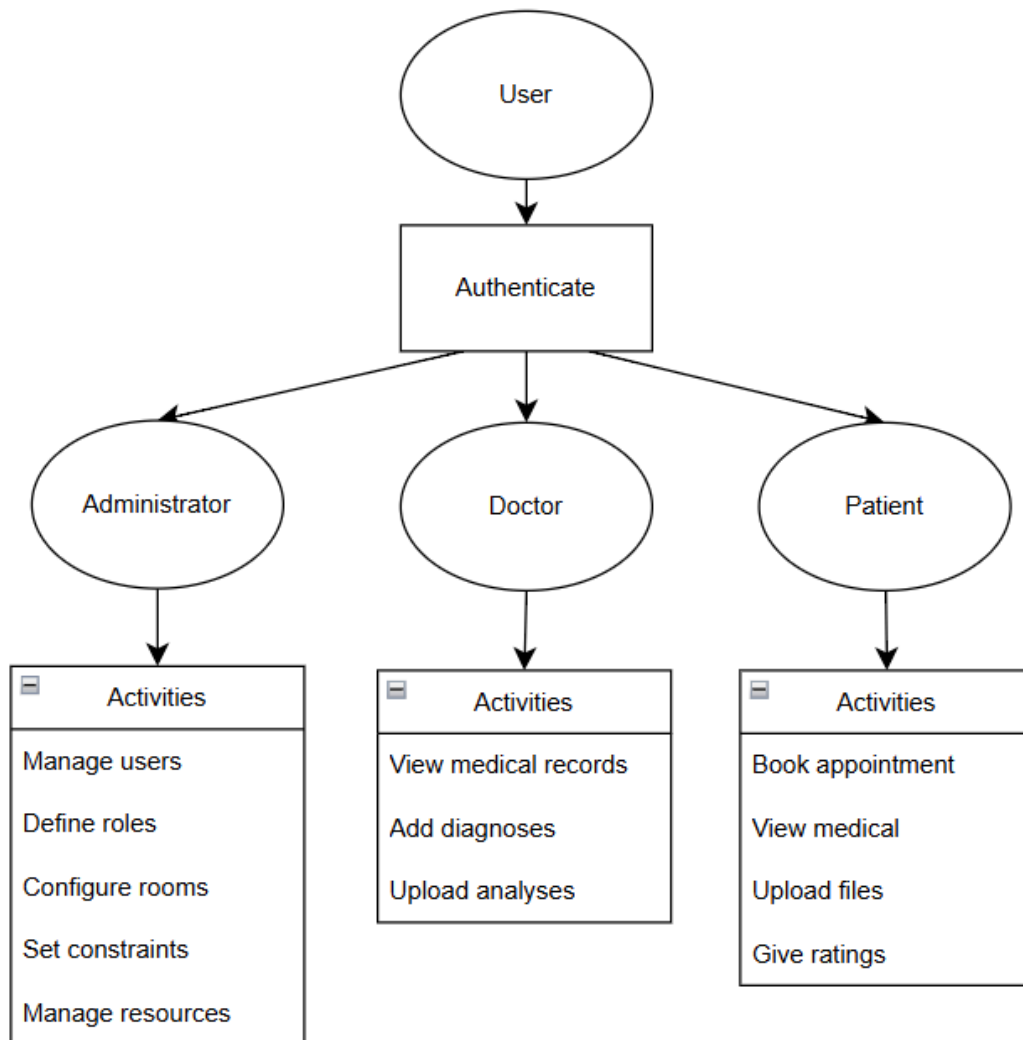
- *Properties:*
 - *id (Unique identifier)*
 - *name (Full name)*
 - *email (Contact email)*
- *Role: Facilitates appointments on behalf of patients and can edit existing appointments. Their presence can be defined as a mandatory constraint during certain consultations or procedures.*

4. Patient Entity

- *Properties:*
 - *id (Unique identifier)*
 - *name (Full name)*
 - *email (Contact email)*
 - *rating (Score from the system)*
 - *medical file (Internal and external records)*
- *Role: Can make online appointments and has the ability to view their own medical file. They give and receive ratings within the clinic's rating system.*

5. Resource Entity

- *Properties:*
 - *id (Unique identifier)*
 - *name (Resource name)*
 - *type (Medical cabinet, diagnostic device, analysis device, procedure room)*
 - *status (Active/Inactive)*
- *Role: Represents the physical assets of the clinic. They are allocated in a calendar for appointments and can be activated or deactivated based on their availability.*



2.4 Operating Environment

Hardware requirements: -Any internet-connected device — desktop PCs, laptops, tablets, or smart phones.

Software requirements: -Operating system: Windows 10 or greater

2.5 Design and Implementation Constraints

The Clinic Management System will be implemented as a web application with a responsive design. Patient data must be stored securely in a relational database with encryption and comply with healthcare privacy regulations. Patient data and medical records must be stored securely in a relational database, encrypted at rest and in transit, and comply with healthcare privacy regulations.

The system will use role-based access control and secure authentication for all users. Interfaces must follow a consistent layout and the clinic's branding guidelines. The codebase should be maintainable and documented, allowing future updates and potential integration with external diagnostic or laboratory systems.

2.6 User Documentation

Not applicable.

2.7 Assumptions and Dependencies

The successful development and deployment of the medical clinic management system rely on several key assumptions and external dependencies.

Assumptions:

- **Digital Literacy:** *It is assumed that the clinic's medical staff (doctors, nurses, administrators) and the patients possess basic digital literacy to navigate the platform, view medical files , and use the online scheduling system.*
- **Hardware Availability:** *We assume the clinic is equipped with the necessary hardware infrastructure (computers, tablets) in medical cabinets and procedure rooms to allow staff to access the system in real-time.*
- **Continuous Operations:** *It is assumed that the system will run in an environment with a stable and continuous internet connection, which is critical for the real-time synchronization of the online appointment calendar.*

Dependencies:

- **Third-Party Software Components:** *The project depends on specific commercial or open-source components, such as SQL Server for database management and potentially libraries like iTextSharp for exporting patient medical records or prescriptions.*
- **Data Privacy Regulations:** *Because the system stores highly sensitive personal and medical data, including internal analyses and external treatments received previously, its implementation is strictly dependent on complying with data protection laws (e.g., GDPR).*
- **External File Formats:** *The feature allowing the addition of external resources regarding patients (investigations, analyses) depends on the assumption that these external documents can be uploaded in standard formats (e.g., PDF, JPEG).*

3. External Interface Requirements

3.1 User Interfaces

*The **Clinic Management System** is designed as a web-based application that allows users to access the platform through standard web browsers.*

GUI Standards and Design Guidelines

To provide a clear and efficient user experience, the interface follows these guidelines:

- **Responsive Design:** *The interface adapts to different screen sizes and devices such as desktops, tablets, and mobile phones.*
- **Consistent Layout:** *All pages follow a similar layout structure to ensure easy navigation and usability.*
- **Clear Navigation:** *The system provides intuitive menus that allow users to easily access major modules such as appointments, medical records, and resources.*
- **Form Design:** *Input forms use clear labels and validation messages to help users enter correct information.*
- **Error Handling:** *The system displays clear and informative error messages when invalid data is entered.*

Standard Buttons and Functions

- **Login / Logout Button:** *Allows users to securely authenticate and exit the system.*
- **Save & Cancel Buttons:** *Allow users to confirm changes or cancel an action when editing data.*
- **Edit / Delete Buttons:** *Allow authorized users to modify or remove information such as appointments or medical records.*

Error Message Display Standards

- **Visual Indicators:** *Incorrect fields are highlighted and accompanied by clear error messages to guide the user in correcting input errors.*

Screen Layout Constraints

- **Minimal Interface Complexity:** *Screens display only the necessary information to avoid clutter.*
- **Role-Based Interface:** *Available features and data are displayed according to the user's role (administrator, doctor, nurse, patient).*
- **Accessibility:** *Frequently used functions such as appointment booking or medical record access are easily accessible.*

1. Login Page

Software Requirements Specification for An informatic system for the management of a Medical Clinic

Page 9

MedicalClinic Register Login

**MediClinic**
Sistem de Management Clinic

Adresă Email

Parolă

☐ Ține-mă conectat Al uitat parola?

Autentificare

Pacient nou? [Înregistrează-te](#)

© 2026 - MedicalClinic - [Privacy](#)

2. Appointment Calendar Page

< > Azi

3 – 9 mai 2026

Luni Săptămână Zi

	dum. 03.05	lun. 04.05	mar. 05.05	mie. 06.05	Joi 07.05	vin. 08.05	sâm. 09.05
07							
08							
09							
10							
11							
12							
13							
14							
15							
16							
17		08:00 - 17:00 CalendarViewing2004@gmail.com - Beana					
18							
19							

3. Admin's dashboard

MediClinic admin@medicalclinic.com Logout

Bun venit, Administrator!
Gestionează resursele, personalul și configurațiile clinicii.

GESTIONARE RESURSE

Specialități

Săli Medicale

Echipamente

Tipuri Consultanțe

GESTIONARE PERSONAL & PACIENȚI

Personal

Specialități Doctori

Dependințe Asistentă-Doctor

Moderare Recenzii

Pacienți

MONITORIZARE

Calendar Programări

Audit Log

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3.2 Hardware Interfaces

Not applicable.

3.3 Software Interfaces

For the medical clinic management system, the connections with other software components include:

Database Management System (DBMS)

- **Software Component:** Microsoft SQL Server
- **Purpose:** Microsoft SQL Server will serve as the primary database management system for storing and managing all clinic data. This includes core roles (doctor, nurse, administrator, patient), defined resources (medical rooms, diagnostic devices, analysis equipment, procedure rooms), appointments, and patient medical records. It integrates seamlessly with ASP.NET Core applications developed in Visual Studio 2022, ensuring efficient data access, security, and scalability.

Web Development Framework (Backend)

- **Software Component:** ASP.NET Core (developed via Visual Studio 2026)
- **Purpose:** This framework will be used for server-side scripting to handle HTTP requests, enforce complex scheduling constraints, process online appointments, and interact securely with the SQL Server database. It also enables the use of Razor Pages for building dynamic web interfaces.

Frontend Framework

- **Software Component:** Razor Pages (ASP.NET Core), HTML, CSS, JavaScript (optional)
- **Purpose:** Razor Pages will be used to build the user interface by combining HTML markup with server-side C# logic. HTML defines the structure of the pages, CSS is used for styling and responsive design, and JavaScript provides additional interactivity when needed. This approach allows doctors and patients to view medical records, nurses to manage appointments, and administrators to control clinic resources efficiently.

Authentication and Authorization

- **Software Component:** ASP.NET Core Identity
- **Purpose:** ASP.NET Core Identity will be used to manage user authentication and authorization. It will ensure that only authorized users can access specific features of the system based on their assigned roles (administrator, doctor, nurse, or patient). The system will use secure login mechanisms and role-based access control to protect sensitive medical information.

Document Processing Library

- **Software Component:** *iTextSharp (version 5.5.13)*
- **Purpose:** *This library will be integrated to generate, export, and format printable medical documents, such as the patient's medical file or reports regarding their medical evolution based on conducted analyses and investigations.*

Communication Protocols

- **Software Component:** *HTTP/HTTPS*
- **Purpose:** *Secure HTTP/HTTPS protocols will be used for communication between the user's web browser and the server, facilitating the exchange of requests and responses for accessing appointments, submitting ratings , or uploading external resources.*

3.4 Communications Interfaces

Web Browser Communication and Electronic Forms Submission:

- **Protocol:** *HTTP/HTTPS*
- **Requirements:** *The system should support communication with web browsers for user interaction through a secure web-based interface. It includes sending and receiving HTTP/HTTPS requests and responses for accessing the centralized calendar and viewing the patient's medical file. Users must be able to securely submit electronic forms through the web interface for tasks such as creating online appointments , adding or editing external medical resources like prior treatments and investigations , and submitting ratings within the system. Because the system processes highly sensitive health information, all form submissions, data transfers, and session management must be strictly handled over encrypted HTTPS connections to ensure data privacy and integrity.*

Email Notifications (Optional but recommended):

- **Protocol:** *SMTP (Simple Mail Transfer Protocol)*
- **Requirements:** *The system should be able to send automated email notifications to users. This includes sending confirmations for online appointments made by patients or nurses, notifying users of appointment updates or cancellations, and alerting patients when a doctor updates their medical file with new information based on recent analyses.*

4. System Features

4.1 User Authentication and Account Management

4.1.1 Description and Priority

Priority: *High*

Description:

Users can authenticate in the system using email and password credentials. The system

manages user accounts, roles, and permissions for different types of users including administrators, doctors, nurses/secretaries, maintenance staff, and patients. The module also maintains an audit log of critical actions performed by users.

4.1.2 Stimulus/Response Sequences

The user enters their email and password in the login fields and presses the login button. The system verifies the credentials and authenticates the user. If authentication succeeds, the system determines the user's role and redirects them to the appropriate dashboard. If authentication fails, the system displays an error message.

Patients can also register by filling in a registration form. After submitting the form, the system validates the data and sends a verification email or SMS.

Administrators can create, edit, or deactivate accounts for clinic staff.

4.1.3 Functional Requirements

REQ-1: Authentication Process: The system shall allow secure login and logout using email and password authentication.

REQ-2: Role Management: The system shall support five user roles: Administrator, Doctor, Nurse/Secretary, Maintenance Staff, and Patient.

REQ-3: Staff Account Management: The system shall allow administrators to create, edit, and deactivate accounts for medical and auxiliary staff.

REQ-4: Patient Registration: The system shall allow patients to create their own accounts through an online registration form.

REQ-5: Account Verification: The system shall verify newly created accounts through email or SMS confirmation.

REQ-6: Password Reset: The system shall allow users to reset their password using a secure link sent to their email address.

REQ-7: Medical Specialty Assignment: The system shall allow associating a doctor with one or more medical specialties.

REQ-8: Staff Dependency Management: The system shall allow defining relationships between staff members (for example assigning a nurse to a specific doctor or department).

REQ-9: Account Recovery: The system shall allow users to recover their account via email or phone number and verify their identity before resetting credentials.

REQ-10: Audit Log Details: The audit log shall store information about the user, the performed action, and the timestamp.

4.2 Resource Management (Rooms and Equipment)

4.2.1 Description and Priority

Priority: High

Description:

The system allows administrators to define and manage physical resources within the clinic such as rooms and medical equipment. Each resource can have different states and can be associated with specific locations.

4.2.2 Stimulus/Response Sequences

An administrator accesses the resource management section.

The administrator can create new rooms or medical equipment entries.

The administrator can update the status of resources.

When a resource becomes unavailable, the system blocks any related bookings.

4.2.3 Functional Requirements

REQ-11: Room Definition: The system shall allow defining and naming rooms within the clinic (e.g., consultation room, operating room, therapy room).

REQ-12: Equipment Definition: The system shall allow defining and naming medical equipment (e.g., ultrasound machine, MRI scanner, ECG device).

REQ-13: Equipment Location Assignment: The system shall allow assigning equipment to specific rooms.

REQ-14: Room Status Management: The system shall allow changing the status of a room (Available, Unavailable) and block bookings during unavailable periods.

REQ-15: Equipment Status Management: The system shall allow changing the status of equipment (Functional, In Service, Maintenance) and block bookings that depend on that equipment.

REQ-16: Equipment Fault Notification: The system shall notify administrators when equipment is marked as defective.

REQ-17: Resource Listing: The system shall display all resources in a table view with filters based on status, type, and location.

4.3 Appointment Scheduling and Constraints

4.3.1 Description and Priority

Priority: High

Description:

The appointment scheduling system manages the booking of consultations while respecting constraints related to staff, rooms, and equipment availability.

4.3.2 Stimulus/Response Sequences

A patient or staff member selects a medical specialty and doctor.

The system displays available time slots.

The user selects a slot and confirms the booking.

The system validates constraints such as staff availability and resource conflicts before confirming the appointment.

4.3.3 Functional Requirements

REQ-18: Specialty Room Constraint: The system shall allow defining rules where certain specialties can only take place in specific rooms.

REQ-19: Nurse Requirement Rule: The system shall allow defining procedures that require the presence of a nurse.

REQ-20: Equipment Conflict Prevention: The system shall prevent the same equipment from being reserved in multiple rooms simultaneously.

REQ-21: Doctor Scheduling Conflict Prevention: The system shall prevent a doctor from being scheduled in two different rooms at the same time.

REQ-22: Nurse Scheduling Conflict Prevention: The system shall prevent a nurse from being scheduled in two different rooms simultaneously.

4.4 Appointment Calendar and Booking

4.4.1 Description and Priority

Priority: High

Description:

The system provides an interactive calendar where users can view and manage appointments. Patients can create appointments online, while staff members can manage bookings for patients.

4.4.2 Stimulus/Response Sequences

A user accesses the calendar view and selects a time slot.

Patients can create new appointments by selecting a doctor and specialty.

Staff members can schedule appointments for patients over the phone or in person.

The system confirms or rejects the appointment depending on availability.

4.4.3 Functional Requirements

REQ-23: Interactive Calendar: The system shall display an interactive calendar with daily, weekly, and monthly views.

REQ-24: Calendar Filtering: The system shall allow filtering the calendar by resource (room/equipment) or user (doctor).

REQ-25: Online Appointment Booking: The system shall allow logged-in patients to book appointments online.

REQ-26: Staff Appointment Creation: The system shall allow nurses or secretaries to create appointments for patients.

REQ-27: Appointment Editing: The system shall allow staff members or doctors to edit existing appointments.

REQ-28: Appointment Cancellation: The system shall allow canceling appointments with a cancellation reason.

REQ-29: Automatic Room Allocation: The system shall automatically assign an available compatible room when creating a booking.

REQ-30: Consultation Duration Configuration: The system shall allow defining standard consultation durations per specialty.

REQ-31: Appointment Confirmation Notification: The system shall send an automatic email confirmation to the patient after booking.

REQ-32: Appointment Reminder Notification: The system shall send an automatic reminder to the patient 24 hours before the appointment.

4.5 Electronic Health Record (EHR)

4.5.1 Description and Priority

Priority: High

Description:

The Electronic Health Record module allows managing patient medical records including diagnoses, consultation notes, external documents, and health indicators.

4.5.2 Stimulus/Response Sequences

A doctor opens a patient profile.

The system displays the patient's medical history.

Doctors can add notes and diagnoses.

Patients can view their own medical records from their accounts.

4.5.3 Functional Requirements

REQ-33: Patient Medical Record Access: The system shall display the patient's medical record including appointment history, diagnoses, and treatments.

REQ-34: Patient Record Viewing: The system shall allow patients to view their own medical history.

REQ-35: Doctor Notes Entry: The system shall allow doctors to add consultation notes and diagnoses.

REQ-36: External Document Upload: The system shall allow uploading external medical documents such as PDFs or medical images.

REQ-37: Medical Record PDF Export: The system shall allow downloading a PDF version of the patient record.

REQ-38: Document Editing and Deletion: The system shall allow administrators or doctors to edit or delete uploaded documents.

REQ-39: Health Indicator Charts: The system shall provide graphical charts showing the evolution of health indicators.

REQ-40: Triage Data Entry: The system shall allow nurses to enter triage data such as blood pressure, weight, and temperature.

REQ-41: Allergy Alert: The system shall display a visual red flag alert for known patient allergies.

REQ-42: Complete Record Export: The system shall allow exporting the entire patient record as an archive or PDF.

4.6 Feedback, Ratings, and Reporting

4.6.1 Description and Priority

Priority: Medium

Description:

This module allows patients to rate consultations and provide feedback about doctors and clinic services. It also provides reporting capabilities for clinic staff.

4.6.2 Stimulus/Response Sequences

After a consultation is completed, the patient receives a notification requesting a rating.

The patient selects a rating and optionally leaves a comment.
The system updates the doctor's rating statistics and displays them on the doctor's profile.

4.6.3 Functional Requirements

REQ-43: Consultation Rating Request: The system shall notify patients after consultations and request a rating between 1 and 5 stars.

REQ-44: Patient Feedback Submission: The system shall allow patients to leave public or anonymous comments.

REQ-45: Doctor Rating Calculation: The system shall calculate and display the average rating of each doctor.

REQ-46: Consultation Rating Display: The system shall display the average rating per consultation in the doctor's profile.

REQ-47: Review Moderation: The system shall allow administrators to approve or remove patient reviews.

REQ-48: No-Show Marking: The system shall allow doctors or nurses to mark a patient as "No Show".

REQ-49: No-Show Restriction: The system shall automatically block online appointment booking for patients with three "No Show" statuses.

REQ-50: Daily Patient Report: The system shall generate a daily report for the secretary containing all expected patients.

REQ-51: Automatic Database Backup: The system shall allow automatic database backups to ensure the integrity of medical history data

5. Other Nonfunctional Requirements

5.1 Performance Requirements

Response Time: *The system's response time should be below 2 seconds for standard operations (loading user profiles, displaying calendar availability) and below 5 seconds for complex operations (generating medical history reports or filtering large sets of resource constraints).*

Concurrency: *The application must support at least 50 concurrent users (patients booking online, doctors accessing records, receptionists managing schedules) without performance degradation, ensuring constant access during peak clinic hours (08:00 - 18:00).*

Database Performance: *Searching for a patient's medical record by name or ID must execute in under 1 second.*

5.2 Safety Requirements

Data Accuracy: While the software does not directly operate medical machinery, incorrect data representation (e.g., showing the wrong medical record, incorrect dosage history, or missing allergy information) can lead to severe patient harm.

Allergy & Risk Warnings: The system must visually highlight (e.g., a mandatory "red flag" banner) any critical patient allergies or severe conditions on the doctor's dashboard before any consultation begins.

Data Loss Prevention: To prevent the loss of critical health records, the system must perform automated daily backups and ensure transactional integrity (e.g., if a system crash occurs during a booking, the calendar must not show phantom appointments).

5.3 Security Requirements

Access Control: Only authorized personnel can access specific modules (e.g., patients can only view their own data; doctors can view/edit records of their assigned patients; maintenance staff can only view equipment status).

Authentication & Password Policy: Passwords must be at least 10 characters long and include at least 1 special character, 1 number, 1 uppercase letter, and 1 lowercase letter. Failed login attempts will lock the account after 5 tries.

Data Encryption: Sensitive medical and personal data (Electronic Health Records) must undergo AES-256 encryption both at rest (in the database) and in transit (via TLS 1.3/HTTPS).

5.4 Software Quality Attributes

1. There shall be no unreachable code.

All code written must be executable and necessary for the application.

2. Meaningful variable and method names must be used.

Names should clearly describe the purpose of the variable, class, or method.

3. camelCase shall be used for local variables and method parameters.

4. PascalCase shall be used for class names and method names.

5. Code must be properly indented and formatted for readability.

6. Functions and methods should remain small and focused on a single responsibility.

7. Input data must always be validated before being processed by the system.

- 8. Sensitive data such as passwords must never be stored in plain text.**
- 9. Error handling must be implemented for critical operations to prevent system crashes.**
- 10. The code must be documented where necessary to explain complex logic or system behavior.**

5.5 Business Rules

1. **BR-1 (Medical Authority):** Only users with the "Doctor" role can add diagnoses, prescribe treatments, or validate external medical investigations.
2. **BR-2 (Resource Locking):** If a resource (room/equipment) is marked as "Under Maintenance" by the Admin or Maintenance staff, the system will instantly block any new appointments requiring that resource and flag existing ones for rescheduling.
3. **BR-3 (Appointment Cancellation):** Patients can only cancel their appointments online up to 24 hours before the scheduled time. Cancellations within 24 hours require a phone call to the clinic.
4. **BR-4 (No-Show Policy):** If a patient accumulates 3 "No Show" statuses (marked by the doctor or receptionist), their online booking privileges are automatically suspended until reviewed by an administrator.
5. **BR-5 (Constraint Precedence):** A booking cannot be confirmed unless all constraints are met simultaneously (Doctor is available + Room for that specific specialty is available + Required equipment is functional + Required assisting nurse is available).

6. Other Requirements

<Define any other requirements not covered elsewhere in the SRS. This might include database requirements, internationalization requirements, legal requirements, reuse objectives for the project, and so on. Add any new sections that are pertinent to the project.>

Appendix A: Glossary

<Define all the terms necessary to properly interpret the SRS, including acronyms and abbreviations. You may wish to build a separate glossary that spans multiple projects or the entire organization, and just include terms specific to a single project in each SRS.>

Appendix B: Analysis Models

<Optionally, include any pertinent analysis models, such as data flow diagrams, class diagrams, state-transition diagrams, or entity-relationship diagrams.>

Appendix C: To Be Determined List

<Collect a numbered list of the TBD (to be determined) references that remain in the SRS so they can be tracked to closure.>